IB MATH AA SL

Lesson 36

Tangents, Normals & Rates of Change

Calculus · 60-minute lesson

Syllabus SL 5.4 ·

Do Now — recall

ON YOUR WHITEBOARD

1. $y = x^2$. Find dy/dx.
2. Gradient of $y = x^2$ at $x = 3$?
3. Equation of a line through (1,2) gradient 4?

Answers (click to reveal)

1. $2x$
2. 6
3. $y - 2 = 4(x - 1)$

What we're learning today

LEARNING OBJECTIVES

1. **Find**
   the gradient of a tangent at a point.
2. **Find**
   the equation of a tangent line.
3. **Find**
   the equation of a normal line.
4. **Apply**
   derivatives as rates of change.

SUCCESS CRITERIA — I CAN...

✓ use $f'(a)$ for the tangent gradient at $x = a$

✓ write a tangent's equation with $y - y_1 = m(x - x_1)$

✓ use gradient $-1/f'(a)$ for the normal

✓ interpret a derivative as a rate of change

Tangents

GRADIENT = f′(a)

$$y - y_1 = m(x - x_1), \quad m = f'(a)$$

Step 1
differentiate to get $f'(x)$

Step 2
find $m = f'(a)$ at the point

Step 3
use $y - y_1 = m(x - x_1)$

NEED A POINT

Find $y_1 = f(a)$ by substituting into the curve.

Then
you have (x_1, y_1) and m.

Equation of a tangent

QUESTION

Find the tangent to $y = x^2$ at the point where $x = 3$.

SOLUTION

Point: $y = 3^2 = 9$, so $(3, 9)$.

Gradient: $dy/dx = 2x$, so $m = 2(3) = 6$.

Use $y - y_1 = m(x - x_1)$:

$$y - 9 = 6(x - 3)$$

$$y = 6x - 9$$

Your turn — tangents

ANSWER IN YOUR BOOK

1. $y = x^2$, gradient at $x = 5$?
2. $y = x^3$, gradient at $x = 2$?
3. Tangent gradient where $dy/dx=2x$ at $x=-1$?

Answers (click to reveal)

1. 10
2. 12
3. −2

Normals

PERPENDICULAR TO TANGENT

$$m_{normal} = -\frac{1}{f'(a)}$$

Normal
at right angles to the tangent

Gradient
the negative reciprocal of the tangent's gradient

RATES

dy/dx is a rate of change.

Context
e.g. dV/dt = rate volume changes.

Sign
tells increasing/decreasing.

Equation of a normal

QUESTION

Find the normal to $y = x^2$ at $x = 3$.

(The tangent gradient there is 6.)

SOLUTION

Normal gradient = −1/6. Point (3, 9):

$$m_{normal} = -\frac{1}{6}$$

Use $y - y_1 = m(x - x_1)$:

$$y - 9 = -\frac{1}{6}(x - 3)$$

EXAM-STYLE QUESTION

Exam-style question

Consider $y = x^2 - 4x$.

(a) Find dy/dx. [1]

(b) Find the equation of the tangent at $x = 1$. [4]

MARK SCHEME — reveal after students attempt (tutor only)

(a) $dy/dx = 2x - 4$ (A1)

(b) $m = 2(1)-4 = -2$ (M1); $y = 1-4 = -3$, point $(1,-3)$ (A1); $y+3 = -2(x-1)$ (M1) → $y = -2x - 1$ (A1)

Common mistakes

Need the y-value
Find $y_1 = f(a)$, not just the gradient.

Normal gradient
It is $-1/m$, the negative reciprocal.

Point–slope
Use $y - y_1 = m(x - x_1)$ carefully with signs.

f′ then substitute
Differentiate first, then put in the x-value.

Mixed practice

QUESTIONS

1. $y = x^2$, tangent gradient at $x = 4$?

2. Tangent to $y = x^2$ at $x = 1$ (equation)?

3. Normal gradient if tangent gradient = 2?

4. $y = x^3$, gradient at $x = 1$?

Answers (click to reveal)

1. 8
2. $m=2,(1,1)$: $y=2x-1$
3. $-1/2$
4. 3

Plenary & exit ticket

KEY FORMULAS

$$m = f'(a)$$

$$m_{normal} = -\frac{1}{f'(a)}$$

EXIT TICKET — before you leave

1. $y = x^2$, gradient at $x = 2$?
2. Normal gradient if $m = 4$?
3. Line through (0,1) gradient 3?

Answers (click to reveal)

1. 4
2. $-1/4$
3. $y = 3x + 1$

Self-assess:

- need more help
- nearly there
- confident

Homework

SET ON DR FROST MATHS (auto-marked)

Task:
Tangents, normals & rates.

- Aim for 80%+ before the next lesson
- Re-attempt any flagged questions
- Bring one tricky question to discuss

NEXT LESSON

Lesson 37

Chain, product & quotient rules.

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$